

**Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- Dynamic learning environment that encourages problem-solving using emerging technology
- National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration
- Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si- IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environmen t	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Nikita Dubey



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Session	2025-26(ODD)	Course Name	LAB: Optimization Techniques
Semester	7 th	Course Code	23ADS1707
Roll No	41	Name of Student	Krishna Mishra

Practical Number	2
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To find and classify the stationary points of a multivariable function using partial derivatives and the Hessian Matrix.
Problem Definition	Optimization of Multivariable Functions Using Hessian Matrix
Theory (100 words)	Introduction Optimization is the process of finding the maximum or minimum value of a function. In single-variable optimization, derivatives are used to locate the optimal points efficiently. Theory A multivariable function is a function involving two or more independent variables. Optimization of such functions involves finding the points at which the function attains a local



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maximum, local minimum, or saddle point.

For a function:

the stationary points are obtained by solving:

The nature of the stationary point is determined using the
Hessian Matrix:

Tools Used

- Python 3.x
- Jupyter Notebook / Google Colab
- SymPy Library



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Department of Artificial Intelligence and Data Science



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Procedure and Execution (100 Words)	Algorithm 1. Define the multivariable function. 2. Compute first-order partial derivatives. 3. Equate the partial derivatives to zero. 4. Solve the equations to obtain stationary points. 5. Compute second-order partial derivatives. 6. Form the Hessian Matrix. 7. Calculate the Hessian determinant. 8. Apply Hessian conditions to classify the stationary point. 9. Display the results. Code: <pre>import sympy as sp # Define variables x, y = sp.symbols('x y') # Get user input function_str = input("Enter the two-variable function in terms of x and y\n" "(e.g., x**2 + y**2 - 4*x - 6*y + 13): ") # Convert string to SymPy expression f = sp.sympify(function_str) print("\nThe function is:") print(f) # First partial derivatives fx = sp.diff(f, x) fy = sp.diff(f, y) print("\nFirst partial derivative with respect to x:") print(fx) print("\nFirst partial derivative with respect to y:") print(fy) # Find critical points critical_points = sp.solve([fx, fy], [x, y], dict=True)</pre>
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```
print("\nCritical Points:")
if critical_points:
    for point in critical_points:
        print(point)
else:
    print("No critical points found.")

# Second partial derivatives
fxx = sp.diff(fx, x)
fyy = sp.diff(fy, y)
fxy = sp.diff(fx, y)

print("\nSecond partial derivative fxx:")
print(fxx)

print("\nSecond partial derivative fyy:")
print(fyy)

print("\nMixed second partial derivative fxy:")
print(fxy)

# Classification
print("\nClassification of Critical Points:")

if critical_points:
    for point in critical_points:

        fxx_val = fxx.subs(point)
        fyy_val = fyy.subs(point)
        fxy_val = fxy.subs(point)

        D = fxx_val * fyy_val - (fxy_val ** 2)

        print(f"\nCritical Point: ({point[x]}, {point[y]})")
        print("fxx =", fxx_val)
        print("fyy =", fyy_val)
        print("fxy =", fxy_val)
        print("Hessian Determinant D =", D)

        # Convert to numerical values if possible
        D_num = float(D.evalf())
        fxx_num = float(fxx_val.evalf())

        if D_num > 0 and fxx_num > 0:
            print("→ Local Minimum")
        elif D_num > 0 and fxx_num < 0:
            print("→ Local Maximum")
        elif D_num < 0:
            print("→ Saddle Point")
        else:
            print("→ Test Inconclusive")
else:
```

print("No critical points to classify.")





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Output:

Sample Output

Function:

$x^2 + y^2 - 4x - 6y + 13$

Partial Derivative wrt x:

$2x - 4$

Partial Derivative wrt y:

$2y - 6$



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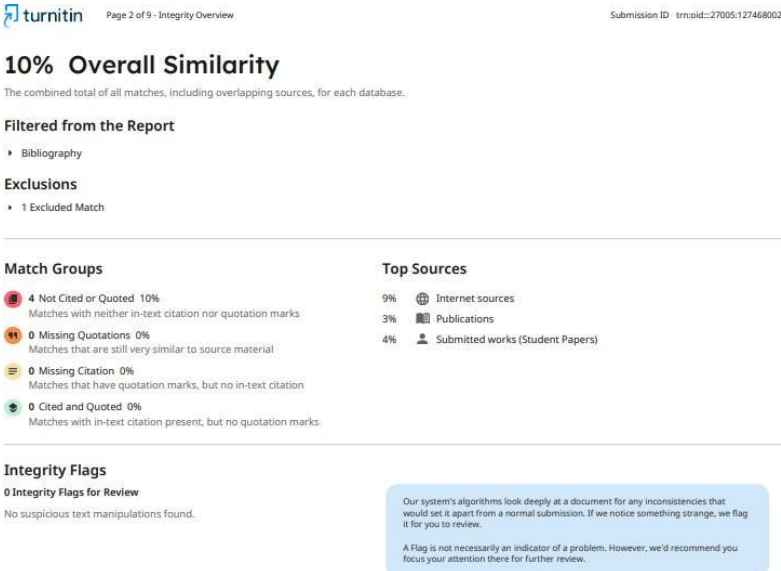
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Output Analysis	The given function is $x^3 - 6x^2 + 9x + 1$. Its critical points are obtained by setting the first derivative equal to zero. Using the second derivative test, $x=1$ is identified as a local maximum and $x=3$ as a local minimum . Thus, the derivative tests successfully determine the optimal points of the function.
Link of student Github profile where lab assignment has been uploaded	https://github.com/nik12345-del/OPTIMIZATION-TECHNIQUES-1/blob/main/optimization%20technique%20-1.ipynb
Conclusion	The first and second derivative tests were successfully used to identify the critical points of the function. The results showed that $x=1$ is a local maximum and $x=3$ is a local minimum, demonstrating the effectiveness of derivative-based optimization.
Plag Report (Similarity index < 12%)	 turnitin Page 2 of 9 - Integrity Overview Submission ID: trnoid:27005:127468002 10% Overall Similarity The combined total of all matches, including overlapping sources, for each database. Filtered from the Report - Bibliography Exclusions 1 Excluded Match Match Groups 4 Not Cited or Quoted 10% Matches with neither in-text citation nor quotation marks 0 Missing Quotations 0% Matches that are still very similar to source material 0 Missing Citation 0% Matches that have quotation marks, but no in-text citation 0 Cited and Quoted 0% Matches with in-text citation present, but no quotation marks Top Sources 9% Internet sources 3% Publications 4% Submitted works (Student Papers) Integrity Flags 0 Integrity Flags for Review No suspicious text manipulations found. Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review. A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.
Date	9/07/2026

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